AI in Robocode

Bot-by-Bot Final Results

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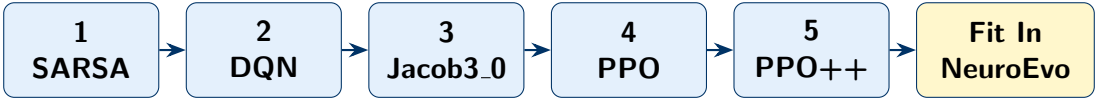
Presentation Order



Talk shape

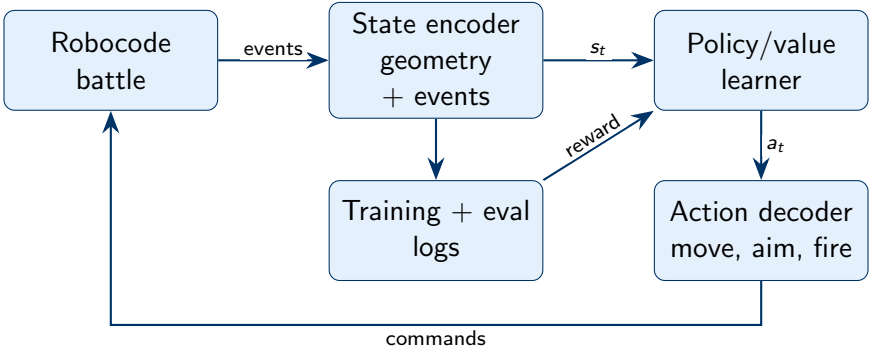
Each bot gets: idea, training evidence, benchmark result, and the one sentence we should remember.

Evidence Corpus

JSONL files	Records	State rows
167	**251,273**	**102,094**
all current structured logs	episodes, summaries, states	battle-state snapshots

- Refreshed analysis: `summary_20260428_091712.json`.
- Structured logs came from training, evaluation, reward-curve, state-collection, PPO curriculum, and stress-test runs.
- Plain `.log` files were used for official Robocode scoreboards and failure notes.

Shared Agent Loop



1. SARSA Bot

On-policy table learning

$$\delta_t = r_t + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t)$$

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha \delta_t$$

- Simple, inspectable baseline.
- Learns from the action it actually takes next.

Scenario	Win	Survival	Damage
vs MeleeDQN	100.0%	289.7	48.2
vs PPOBot	81.6%	188.6	72.8
vs Jacob3_0	43.8%	197.0	63.6
Melee eval	34.7%	370.1	162.7

Remember

SARSA was not the champion, but it was the best non-Jacob melee finisher.

2. DQN Bot

Deep Q-learning target

$$y_t = r_t + \gamma(1 - d_t) \max_{a'} Q_{\bar{\theta}}(s_{t+1}, a')$$

$$\mathcal{L} = \text{SmoothL1}(Q_\theta(s_t, a_t), y_t)$$

- Neural value function instead of a table.
- Replay buffer and target network stabilize updates.
- This implementation struggled to convert state into offense.

Opponent	Win	Reward	Damage
Jacob3_0	2.2%	-8.965	0.0
PPOBot	4.4%	-10.809	2.9
SarsaBot	4.8%	-9.872	0.9
Melee eval	0.0%	-10.954	4.0

Remember

The DQN idea was strong; this bot's learned behavior was not.

3. Jacob3_0

- DQN-based bot with the strongest final benchmark profile.
- Compact state vector: position, energy, walls, gun heat, enemy bearing/distance.
- Best at direct combat and still strong in multi-bot melee.

Opponent	Win	Reward	Surv.	Damage
MeleeDQN	100.0%	0.266	183.4	62.1
PPOBot	92.9%	-3.621	177.9	75.5
SarsaBot	65.7%	-1.557	144.1	66.0
NeuroEvo	100.0%	-1.802	160.6	83.8
Melee eval	60.6%	-0.916	239.5	196.4

Jacob3_0 Map-Size Check

Arena	Win Rate	Avg Reward	Avg Survival
400×400	82.1%	0.925	55.2
800×800	81.0%	0.639	80.5
1200×1200	81.0%	0.723	93.0

Takeaway

The win rate stayed stable as the arena grew, while reward and survival shifted with spacing.

4. PPO Bot

Clipped policy update

$$r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}$$

$$\mathcal{L}^{clip} = \mathbb{E}[\min(r_t A_t, \text{clip}(r_t, 1-\epsilon, 1+\epsilon)A_t)]$$

Scenario	Win	Surv.	Damage
vs MeleeDQN	93.3%	423.7	23.3
vs SarsaBot	22.2%	339.2	39.6
vs Jacob3_0	23.1%	211.7	46.2
Melee eval	6.8%	413.3	92.5

- Direct policy optimization instead of max-Q action selection.
- Often survived for a long time.
- Did not reliably convert survival into wins.

Remember

PPO lasted, but Jacob and SARSA turned fights into wins more often.

PPO Curriculum Run

Late Stage	Win	Survival	Damage
3-bot easy	0.0%	44.0	1.0
4-bot hard	0.0%	25.3	4.8
5-bot mix	0.0%	429.7	5.5
vs Jacob3_0	6.7%	810.2	2.4

What changed on April 28

The newer PPO curriculum completed the staged checkpoint sequence through `stage14`; the logs show more endurance than offense.

5. PPO++

- Repo now includes `PPOBotAdvanced`.
- Separate runtime and agent modules mirror the PPO structure.
- Writer/reader curriculum and staged checkpoints exist through `stage14_ppo_vs_jacob3_0.pt`.

Log status

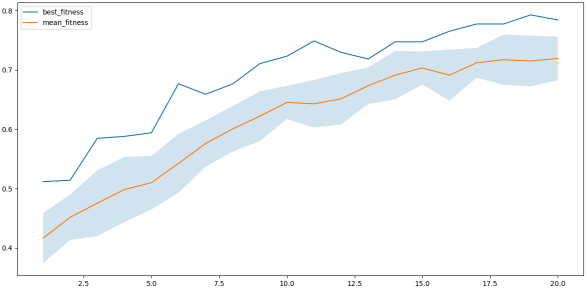
The refreshed log corpus has **0 completed** `PPOBotAdvanced` **summary rows**. PPO++ belongs in the presentation, but not in the ranked benchmark table yet.

Fair line to say

"PPO++ is the upgraded PPO pipeline; code and checkpoints are in place, but the comparable benchmark still needs to be run."

Other Bot That Fits: NeuroEvoMelee

- Evolves a neural policy instead of using gradient learning.
- Fitness improved during evolution.
- Final eval logs show 0.0 damage and 0.0 win rate, so it is not competitive yet.



Fitness signal

$$F = .20S_{survival} + .20S_{placement} + .20S_{damage} + .12S_{kills} + .14W + .14S_{avoided}$$

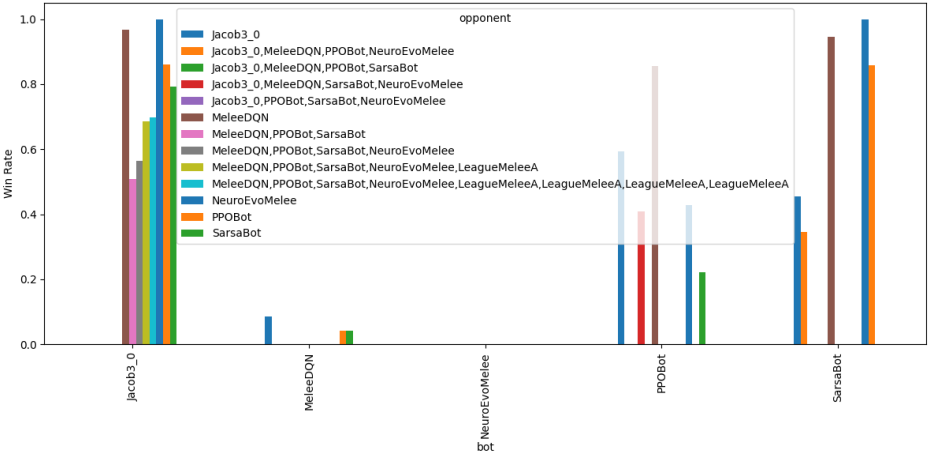
Official Melee Scoreboard

Bot	Rank	Total Score	Firsts	Bullet Damage
Jacob3_0	1	81,613	113	38,858
SarsaBot	2	70,785	68	33,950
PPOBot	3	37,644	19	19,100
MeleeDQN	4	19,131	0	910
NeuroEvoMelee	5	6,700	0	0

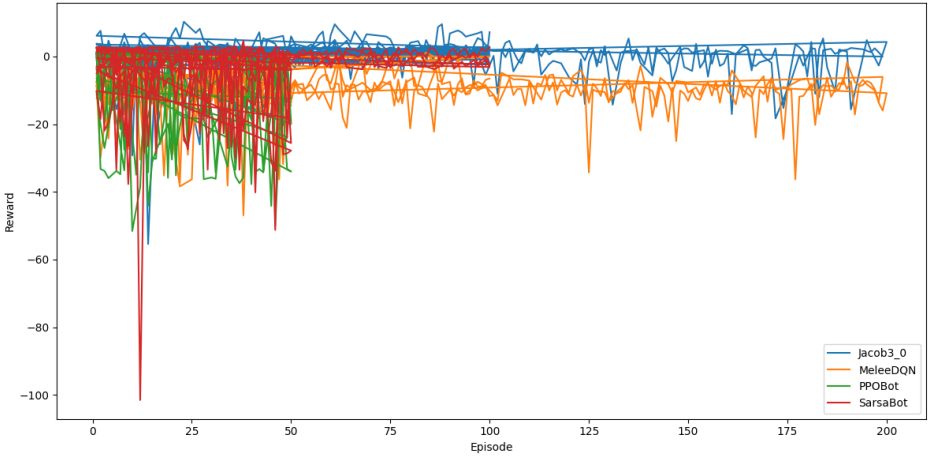
Result

Across the longer logged melee scoring, Jacob3_0 led total score, first-place rounds, and bullet damage.

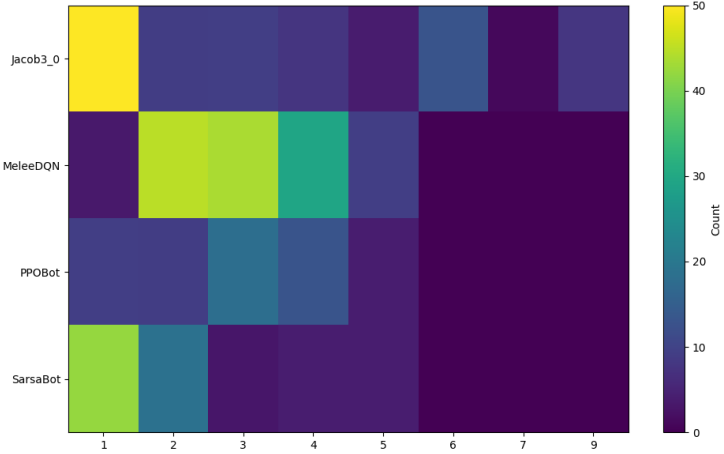
Win Rate Comparison



Reward Curves



Placement Heatmap



What Worked

- Jacob3_0's DQN setup gave the strongest final benchmark result.
- SARSA was a very useful baseline because it stayed competitive without neural-network complexity.
- PPO produced durable behavior, which is a good base for reward and action tuning.
- The log pipeline now captures episodes, summaries, state rows, reward curves, scoreboards, and failure notes.

What Did Not Work Yet

- MeleeDQN learned little offense and collapsed in the final evaluation matrix.
- PPO survived but did not deal enough damage to win consistently.
- PPO++ does not yet have comparable logged benchmark summaries.
- NeuroEvoMelee fitness improved, but final battle behavior still had no reliable damage output.
- The April 28 stress test exposed a PyTorch in-place gradient issue during long training.

Next Changes

Double DQN target

$$y_t = r_t + \gamma(1 - d_t) Q_{\bar{\theta}} \left(s_{t+1}, \arg\max_{a'} Q_\theta(s_{t+1}, a') \right)$$

Reward retuning

$$r_t = c_1 D_{dealt} - c_2 D_{taken} - c_3 H_{wall} + c_4 K + c_5 W$$

Tune the constants against a frozen validation suite instead of only judging from one run.

Conclusion

- Ordered story: SARSA baseline, DQN attempt, Jacob3_0 winner, PPO contender, PPO++ next.
- Jacob3_0 finished first in the logged melee scoreboard and dominated most 1v1 evaluations.
- SARSA earned the clearest second-place story; PPO needs more reward/action work.
- The next fair test is to run PPO++ through the exact same benchmark matrix.